HMGP 7622

Rethinking data analysis

A researcher's guide to avoiding missteps and misuse

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Day 08

Measuring associations

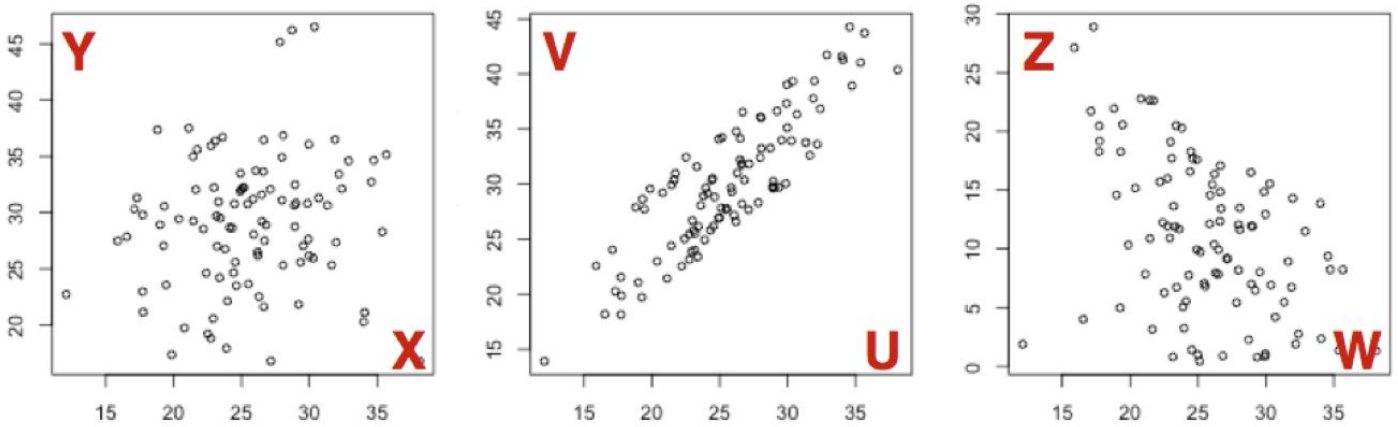
- Calculating correlation
- Limitations
- False positives
- Visual inference

Calculating correlation

Variables

Attributes / Features

x	10	8	13	9	11	14	6	4	12	7	5
y	8.04	6.95	7.58	8.81	8.33	9.96	7.24	4.26	10.84	4.82	5.68



Correlation coefficient

Pearson Correlation Coefficient

- Measures <u>linear</u> relationship between variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

- n is the sample size
- x_i, y_i are the single samples indexed with *i*
- $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ (the sample mean); and analogously for $\bar{y}$

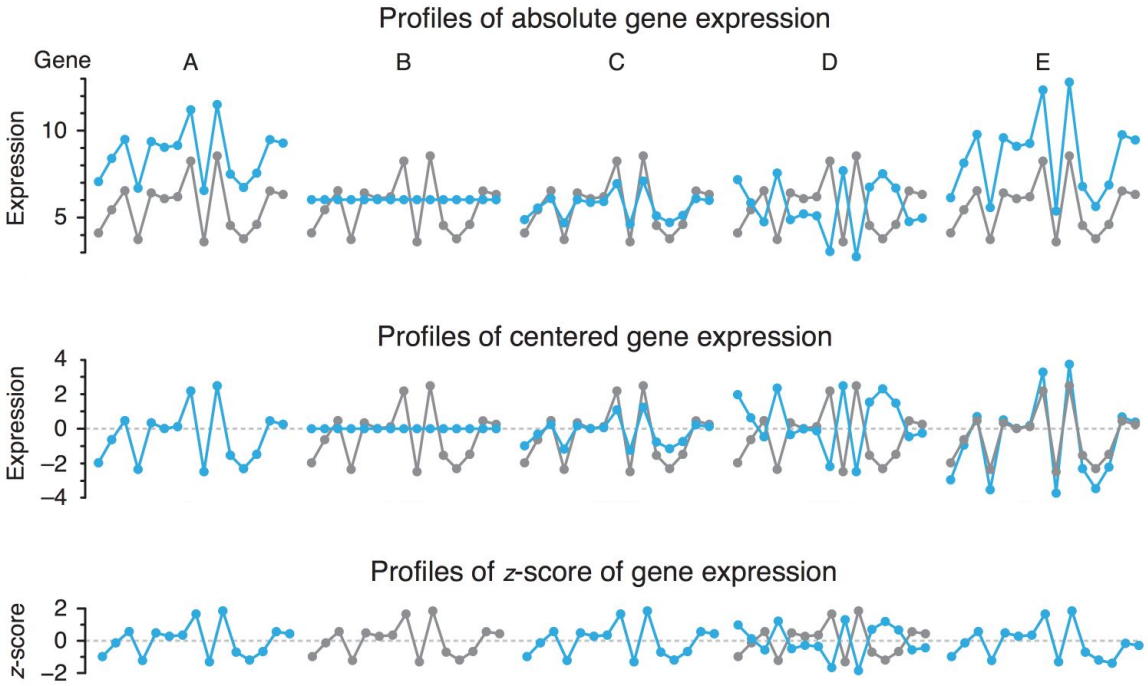
$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

Wikipedia

Correlation coefficient

Pearson Correlation Coefficient

- Captures the relationship between 2 vectors after centering each vector by its mean and scaling by its standard deviation.
- The final quantities for each vector are called z-scores.



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https://www.nature.com/articles/nmeth.4299

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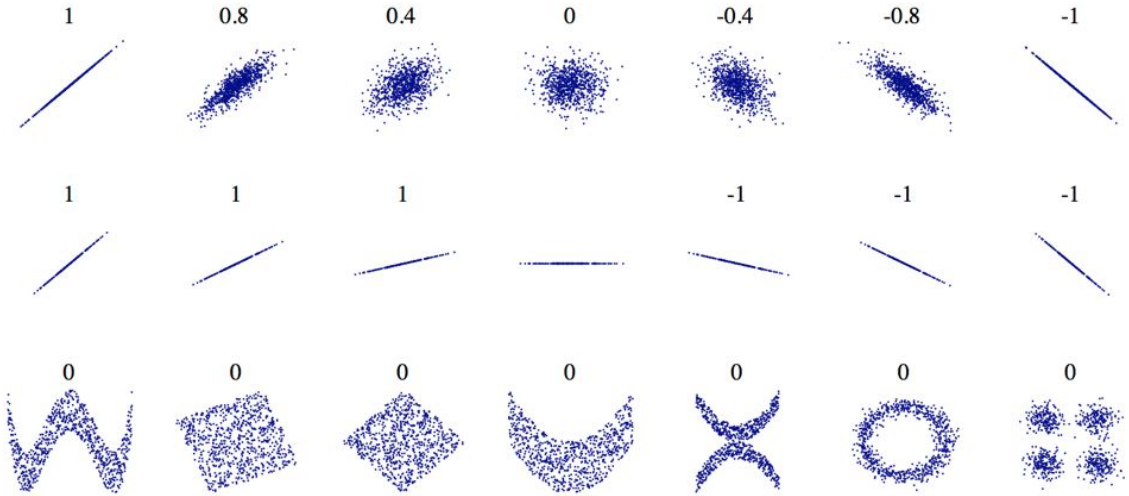


$-1 \leq r \leq +1$

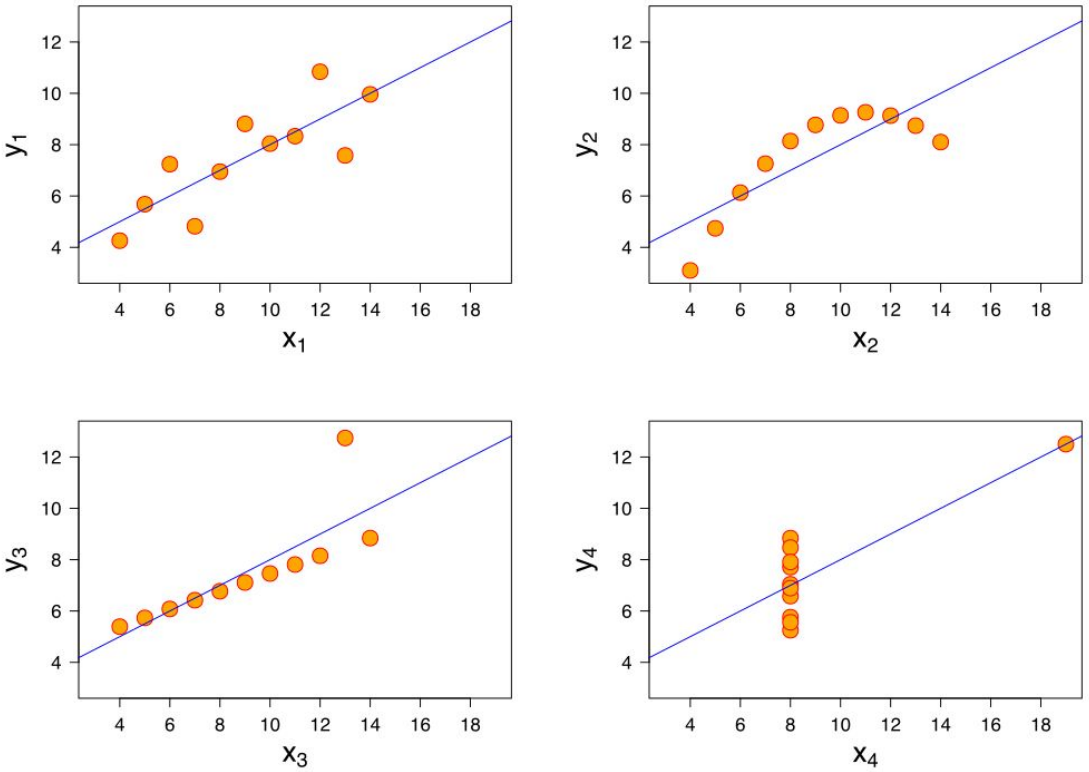
−1 is total −ve correlation | 0 is no correlation | +1 is total +ve correlation

Wikipedia

Anscombe's quartet: "calculation are exact; graphs are rough!"

11 data points

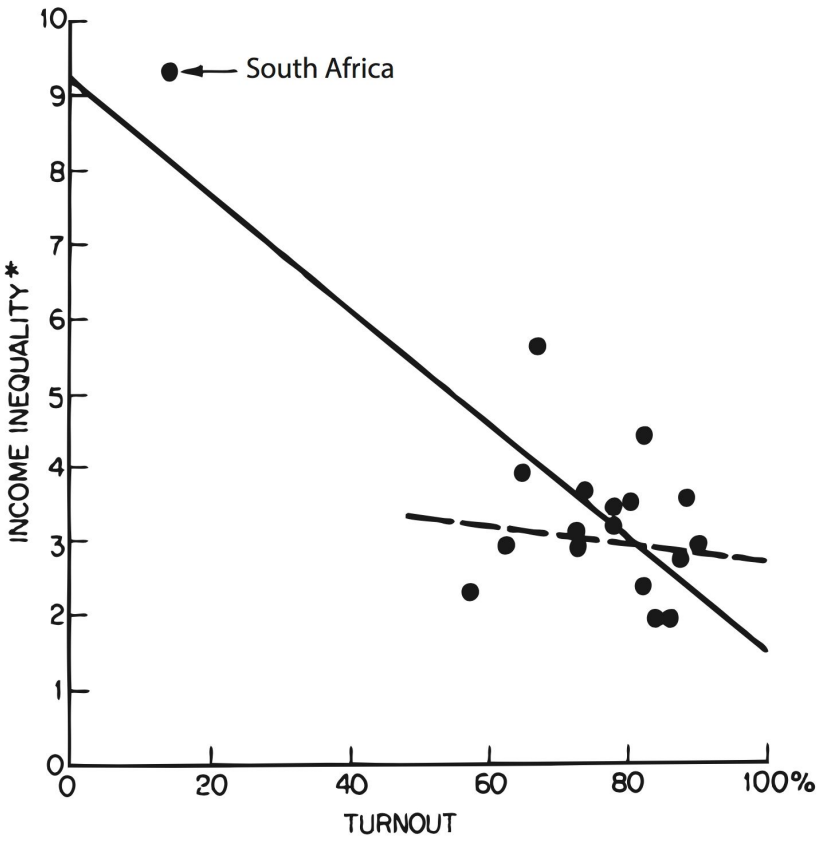
- Mean (x) = 9
- Var (x) = 11
- Mean (y) = 7.50
- Var (y) ~ 4.12
- Cor (x, y) = 0.816
- Linear regression line:
  - y = 3.00 + 0.500x



Anscombe, F. J. (1973). "Graphs in Statistical Analysis". American Statistician 27 (1): 17–21.

Wikipedia

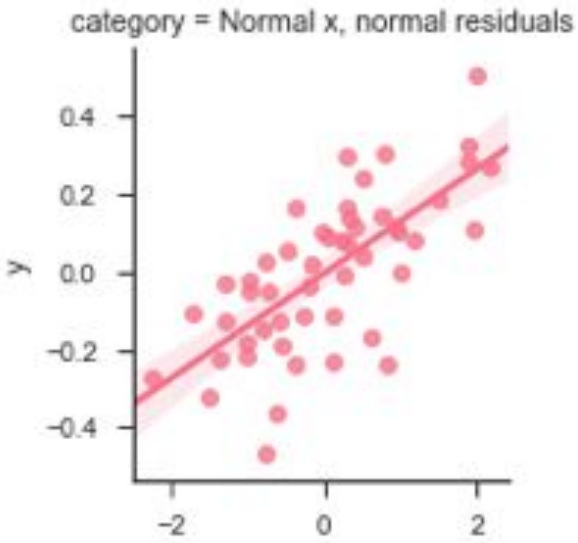
What does a correlation coefficient tell you about the data?



http://socviz.co/lookatdata.html

What does a correlation coefficient tell you about the data?

Correlation = 0.7



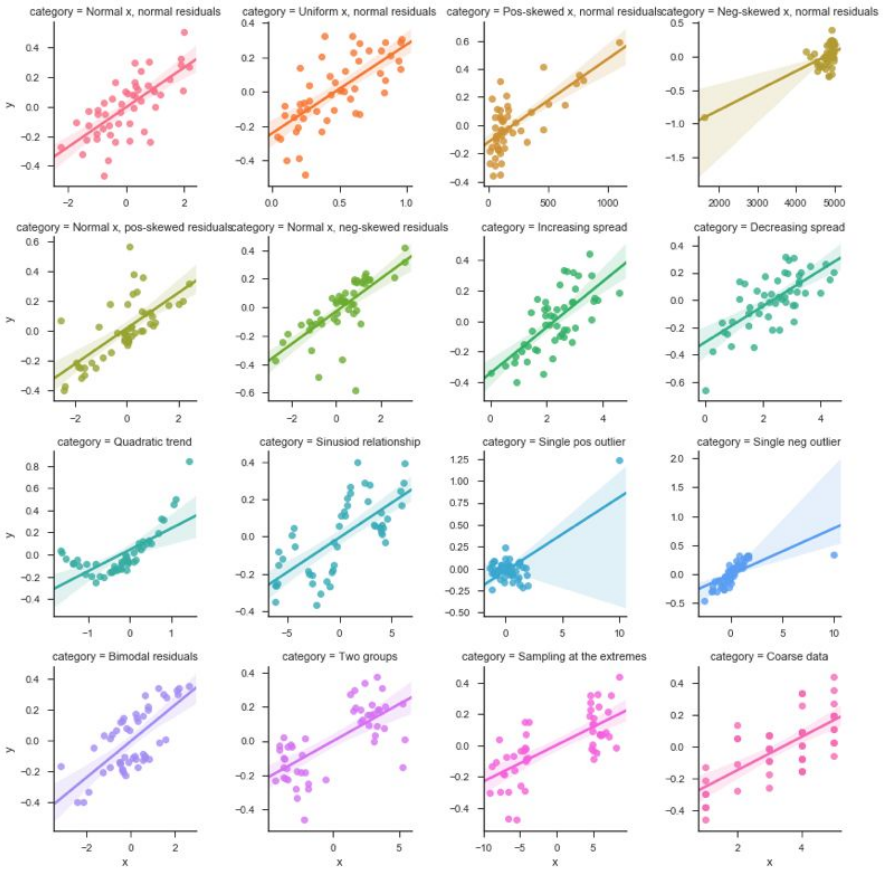
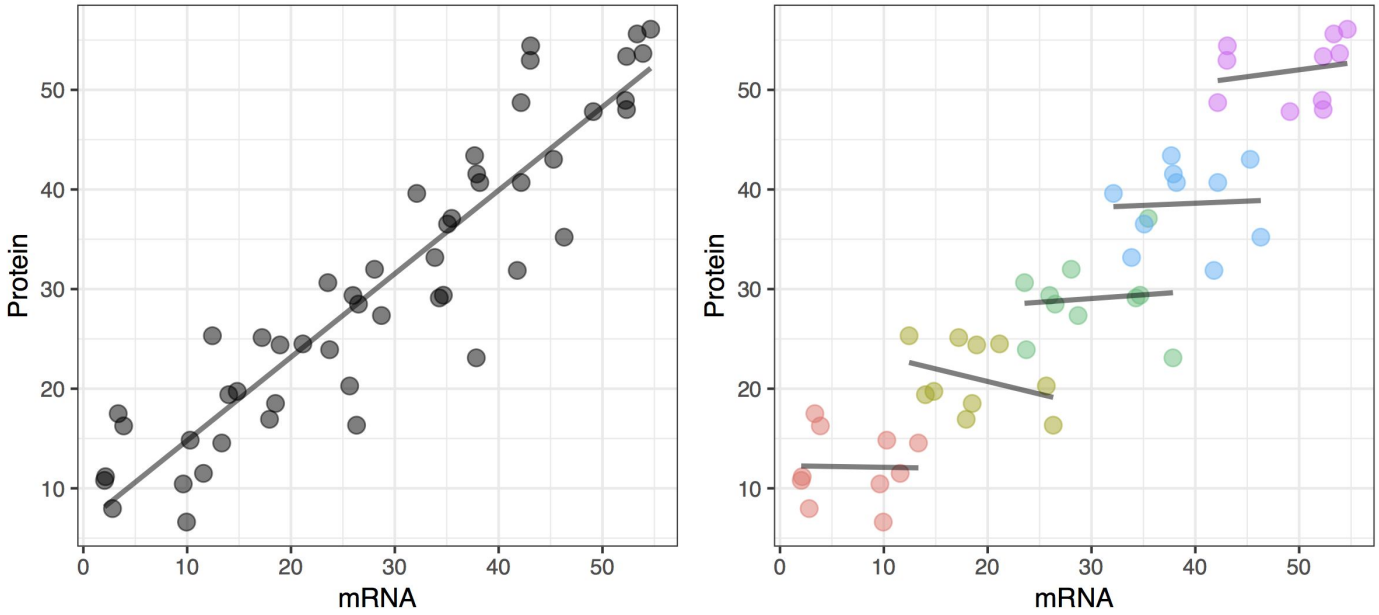


Figure adapted from code by Jan Vanhove

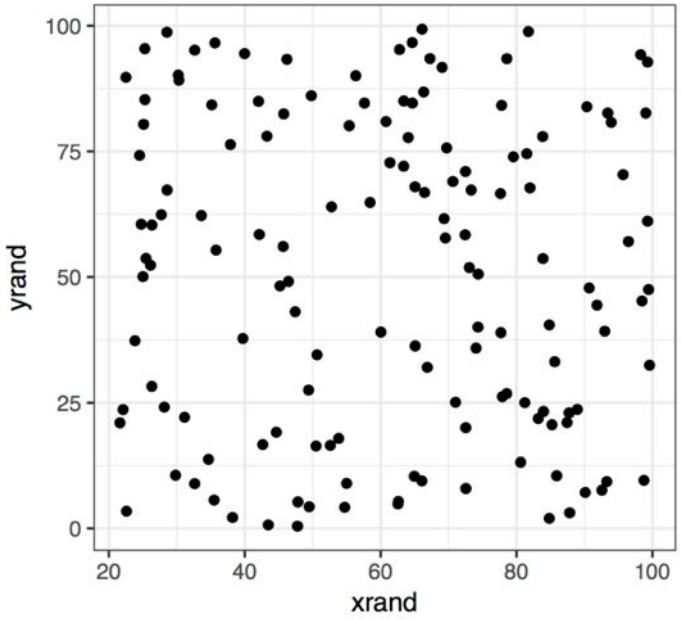
What does a correlation coefficient tell you about the data?

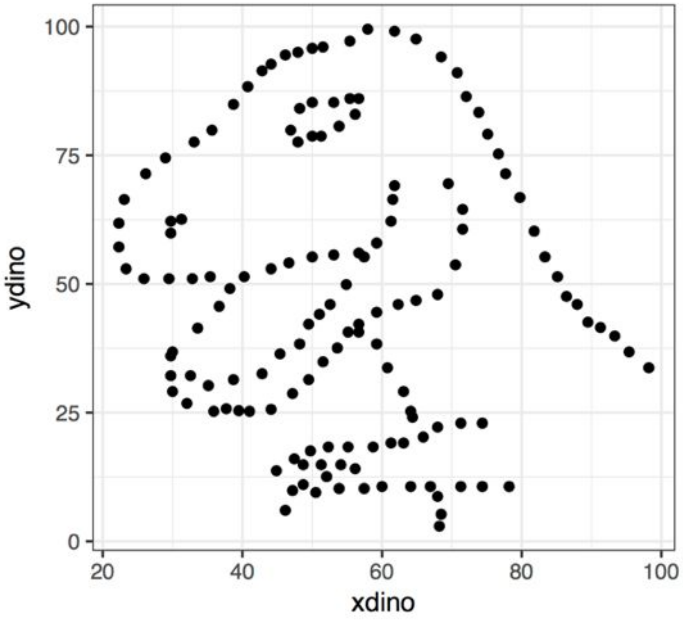
Simpson's Paradox



What does a correlation coefficient tell you about the data?

Correlation ≈ 0





https://www.autodeskresearch.com/publications/samestats

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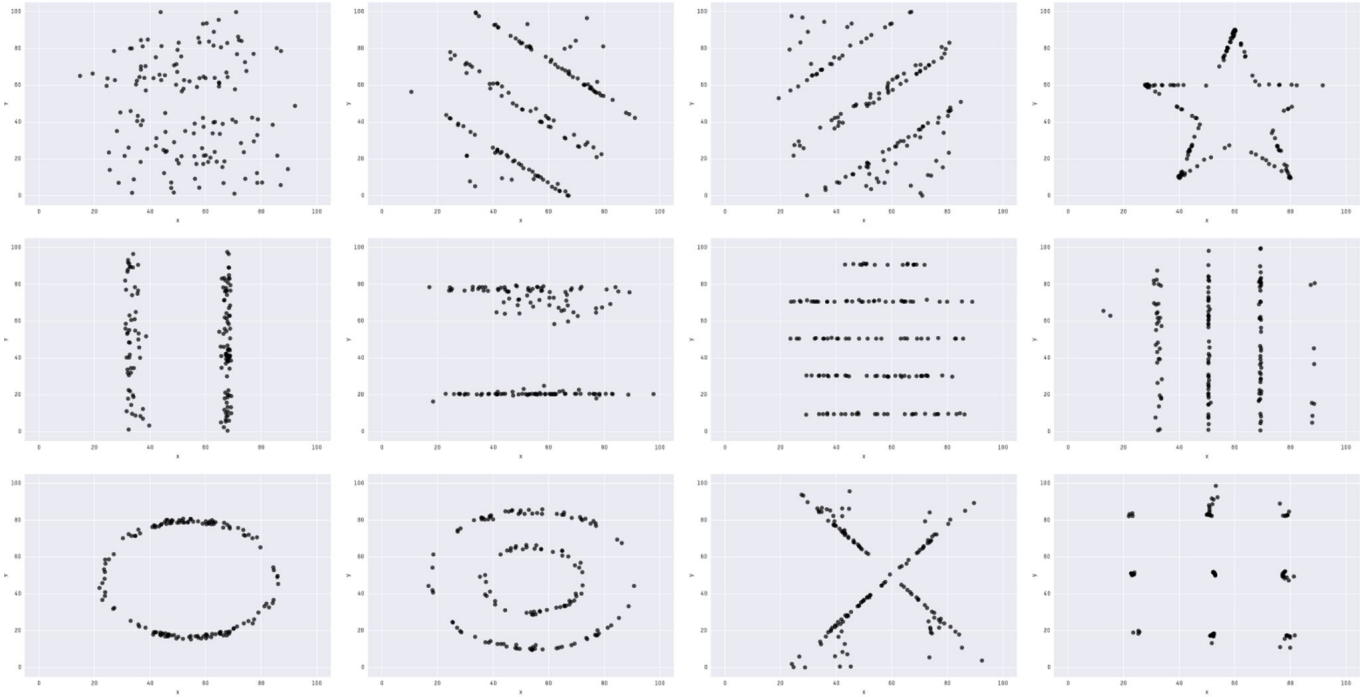

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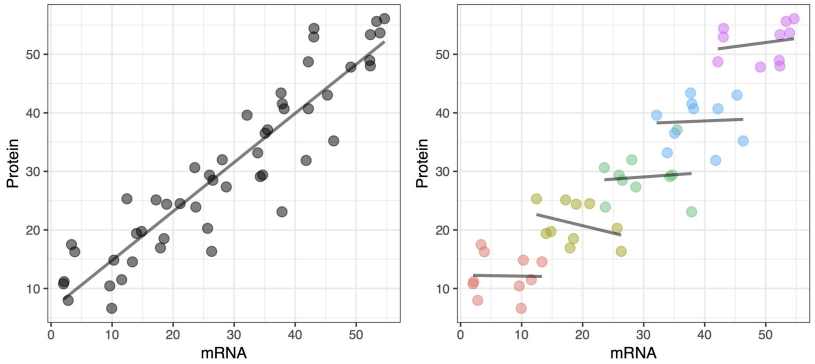
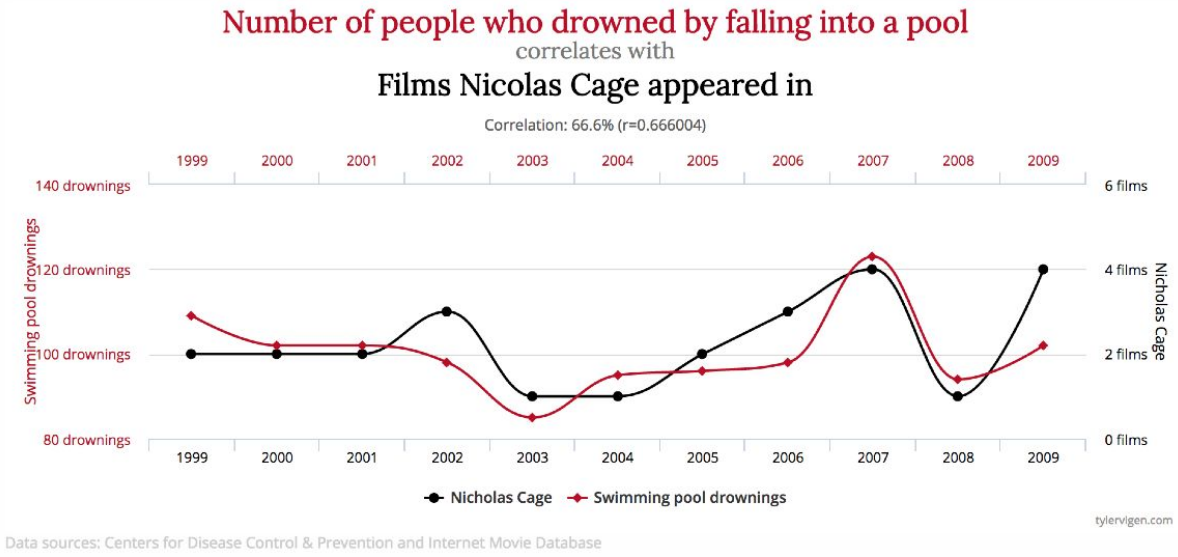


Success rates of kidney stone removal surgeries			
Treatment	**Diameter < 2 cm**	**Dia. ≥ 2 cm**	**Overall**
Open surgery	93%	73%	78%
Percutaneous nephrolithotomy	87%	69%	83%

Spurious correlations

What does Nicholas Cage have to do with people drowning in swimming pools?



Strong correlation b/w annual chocolate consumption & number of Nobel laureates for different countries ($r(20)=.79$; $p<0.001$) → chocolate intake provides nutritional ground for sprouting Nobel laureates

Spurious correlations

Simulate fluctuations in correlation coefficients

- Repeat 10,000: Calculate correlation coefficients of $n = 10$ samples of two independent uniformly distributed variables between (0, 1). Plot a histogram.
- Mark statistically significant coefficients ($\alpha = 0.05$).

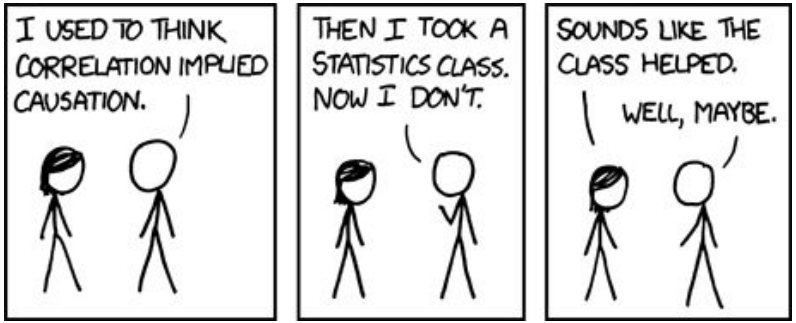
Vary sample size $n = \{5, 10, 20, 50\}$.

- For each, simulate the distribution of correlation coefficients and mark the coefficient corresponding to $\alpha = 0.05$.

Correlation does not imply causation

- Correlation can occur by random chance.
- Confounding variables could lead to correlation.
- Controlling for a “collider” variable could lead to correlation.
- Even when there is causation, there might not be obvious correlation.

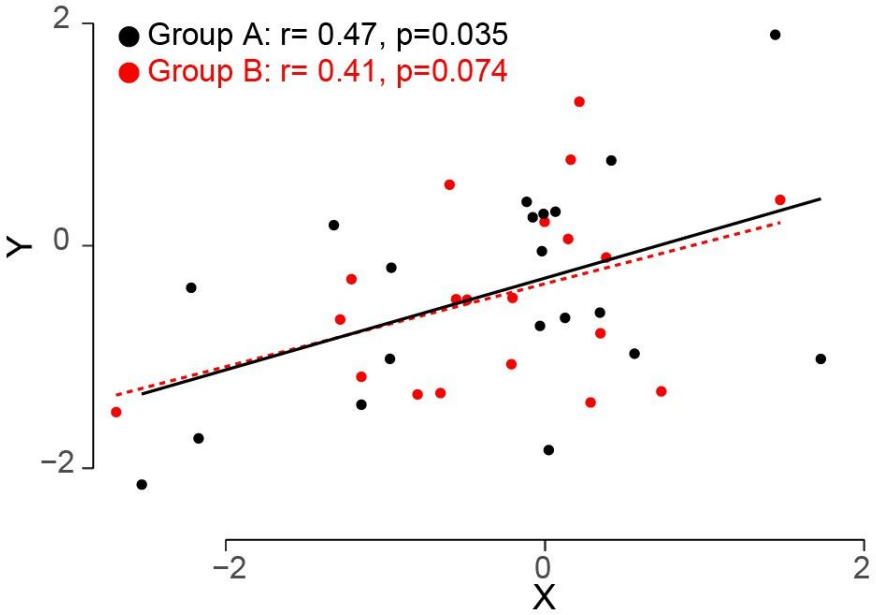
Detecting spurious results requires subject-matter knowledge to both ask if the result defies common sense and perform extremely well-controlled permutation tests.



xkcd.com/552

Interpreting comparison b/w two correlation w/o comparing them

Conclusion regarding the impact of an intervention based on correlation in treatment group vs. correlation in control group.



https://elifesciences.org/articles/48175

Many correlation/distance measures

Pearson Correlation Coefficient

Spearman Rank Correlation

Euclidean Distance

Mutual Information

...

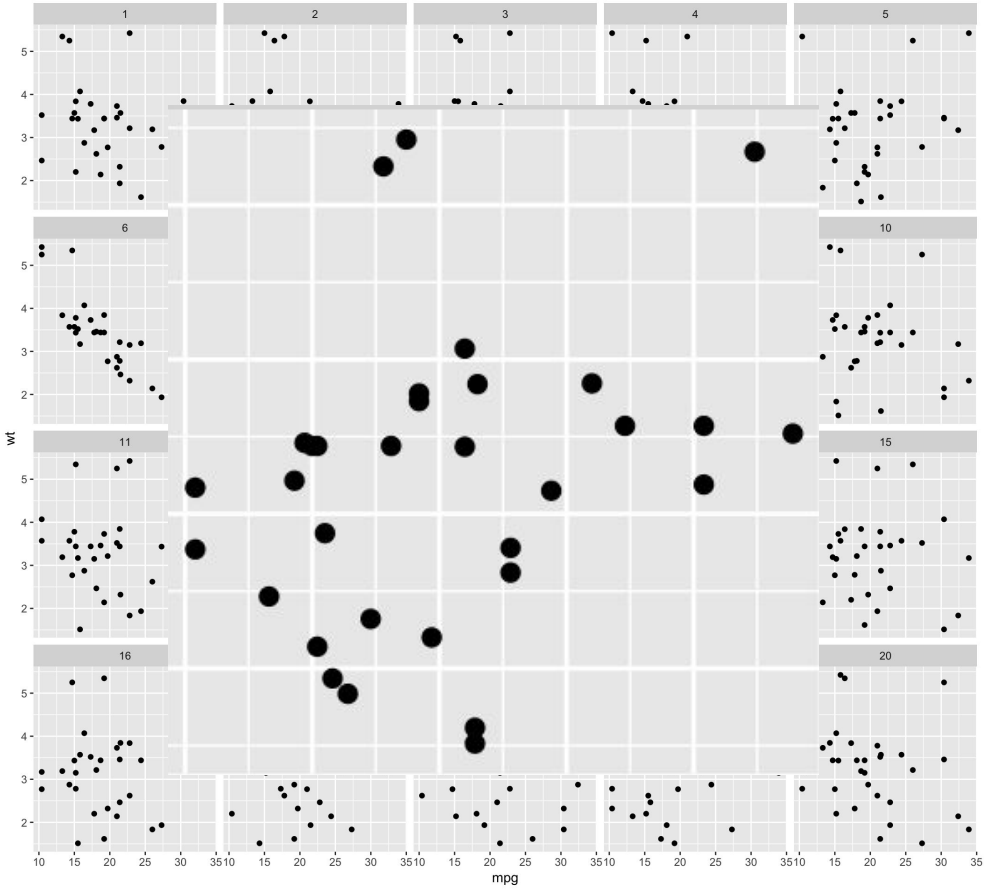
$$d = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

$$r = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right)$$

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} [rank(x_i) - rank(y_i)]}{n(n^2 - 1)}$$

Wikipedia

Spurious correlations – But it *looks* associated!



Create a <u>lineup</u> for visual inference

Place the plot of the real data amongst a set of null plots to create a lineup; Null plots are generated in a way consistent with the null hypothesis.

If the observer can pick the real data as different from the others, this puts weight on the statistical significance of the structure in the plot.

https://lib.dr.iastate.edu/cgi/viewcontent.cgi?article=4995&context=etd

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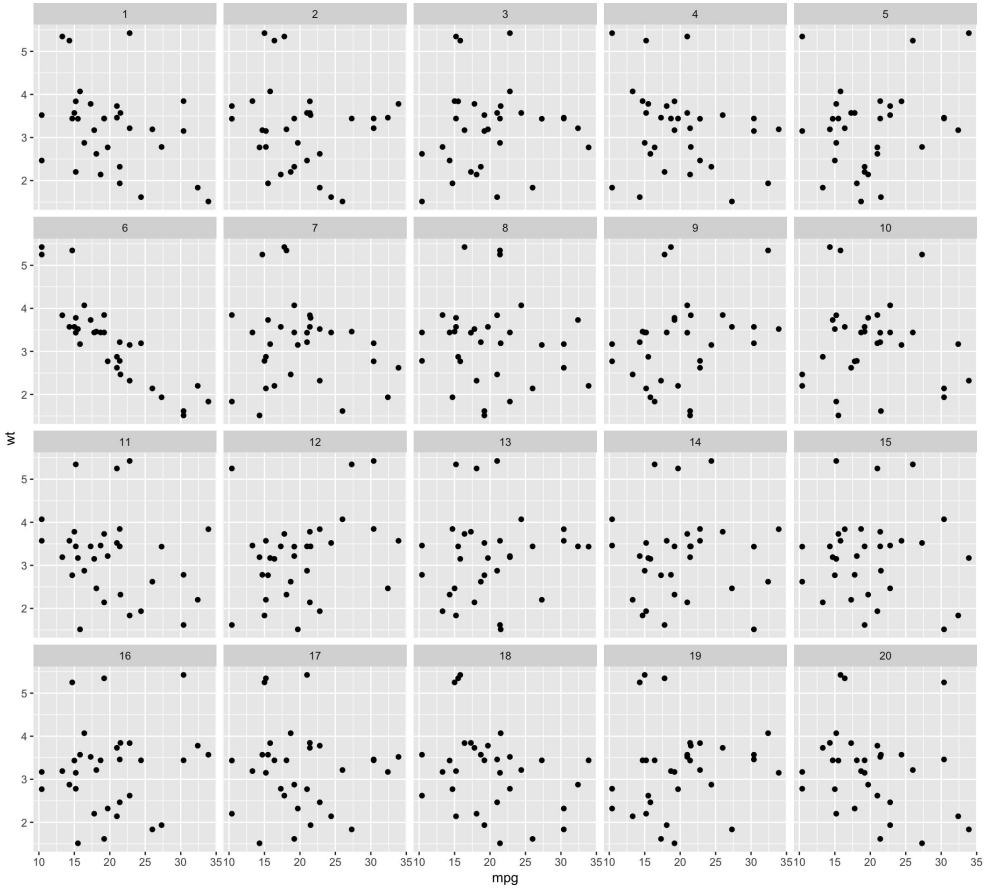

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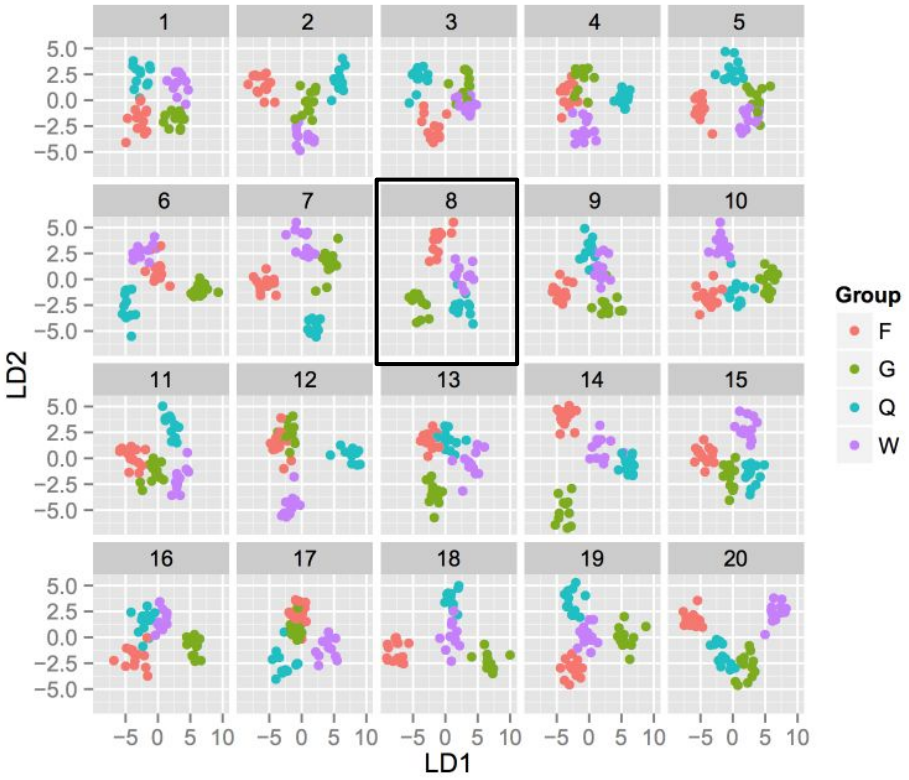

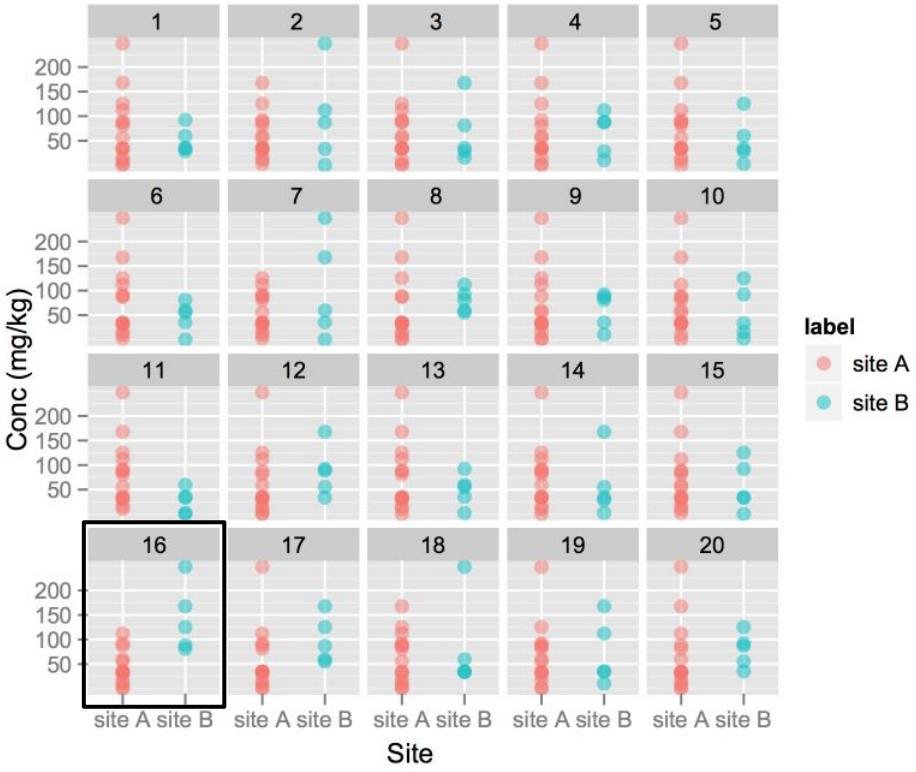

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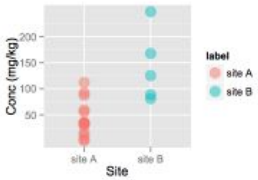
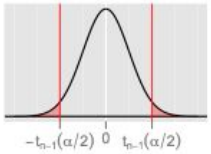
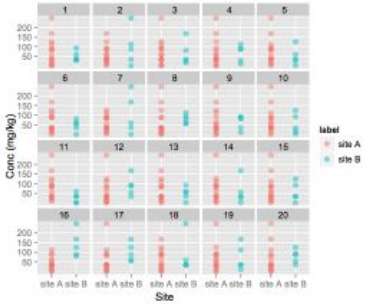
Spurious correlations – But it *looks* associated!

	Mathematical Inference	Visual Inference
Hypothesis	$H_0 : \mu_1 = \mu_2$ vs $H_a : \mu_1 \neq \mu_2$	$H_0 : \mu_1 = \mu_2$ vs $H_a : \mu_1 \neq \mu_2$
	$\downarrow$	$\downarrow$
Test Statistic	$T(y) = \frac{\bar{y}_1 - \bar{y}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$T(y) =$ 
	$\downarrow$	$\downarrow$
Sampling Distribution	$f_{T(y)}(t)$; 	$f_{T(y)}(t)$; 
	$\downarrow$	$\downarrow$
Reject H_0 if	observed T is extreme	observed plot is identifiable